

Toxic Threat Zone

ALOHA® 5.4.7



Time: July 21, 2025 1735 hours PDT (using computer's clock)

Chemical Name: AMMONIA

Wind: 1.55 meters/second from 271° true at 10 meters

THREAT ZONE:

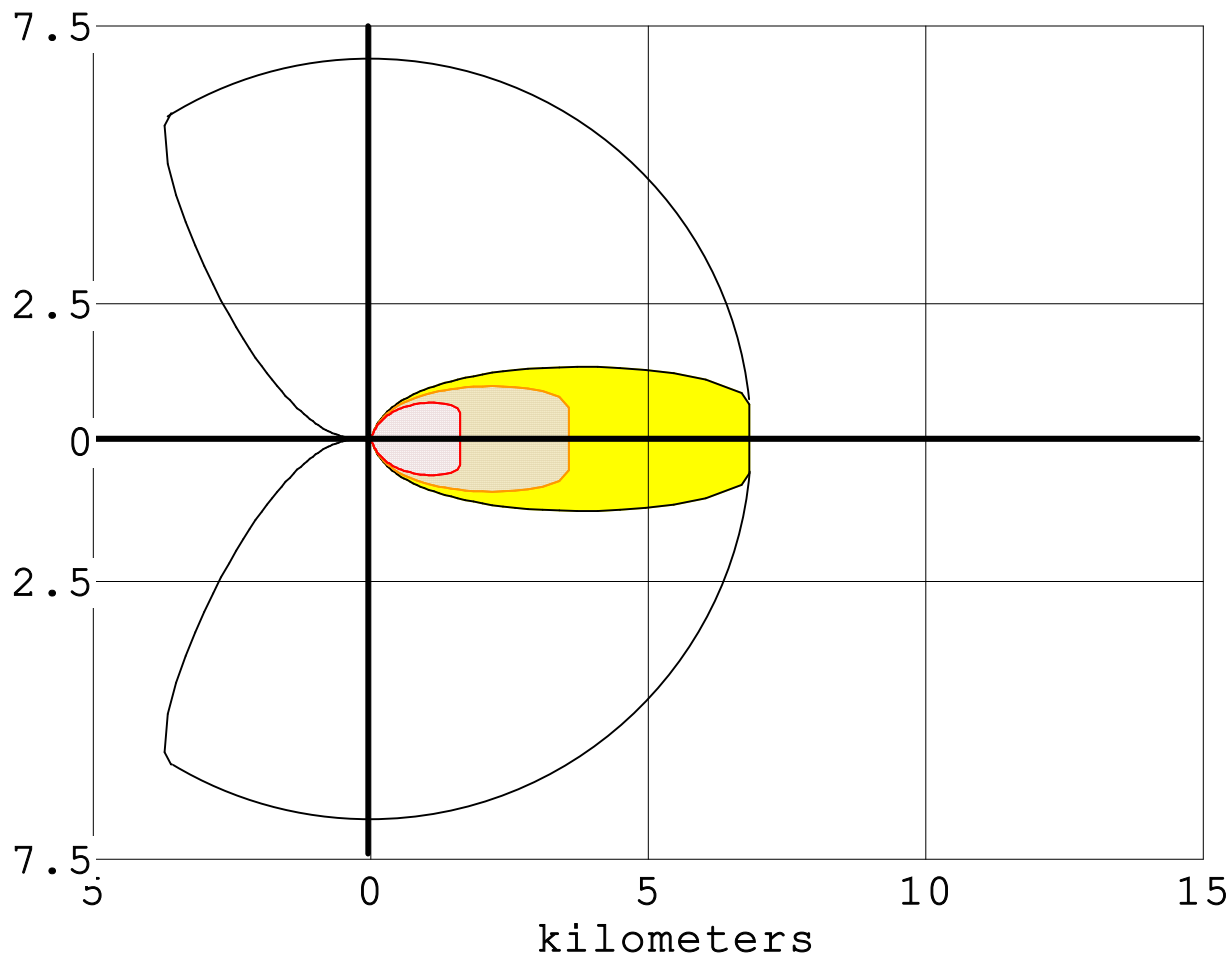
Model Run: Heavy Gas

Red : 1.6 kilometers --- (1100 ppm = AEGL-3 [60 min])

Orange: 3.6 kilometers --- (160 ppm = AEGL-2 [60 min])

Yellow: 6.8 kilometers --- (30 ppm = AEGL-1 [60 min])

kilometers



-  greater than 1100 ppm (AEGL-3 [60 min])
-  greater than 160 ppm (AEGL-2 [60 min])
-  greater than 30 ppm (AEGL-1 [60 min])
-  wind direction confidence lines

Text Summary

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SITE DATA:

Location: TULARE, CALIFORNIA, USA, CALIFORNIA
Building Air Exchanges Per Hour: 0.28 (unsheltered single storied)
Time: July 21, 2025 1735 hours PDT (using computer's clock)

CHEMICAL DATA:

Chemical Name: AMMONIA
CAS Number: 7664-41-7 Molecular Weight: 17.03 g/mol
AEGL-1 (60 min): 30 ppm AEGL-2 (60 min): 160 ppm AEGL-3 (60 min): 1100 ppm
IDLH: 300 ppm LEL: 150000 ppm UEL: 280000 ppm
Ambient Boiling Point: -33.6° C
Vapor Pressure at Ambient Temperature: greater than 1 atm
Ambient Saturation Concentration: 1,000,000 ppm or 100.0%

ATMOSPHERIC DATA: (MANUAL INPUT OF DATA)

Wind: 1.55 meters/second from 271° true at 10 meters
Ground Roughness: open country Cloud Cover: 4 tenths
Air Temperature: 18.54° C
Stability Class: D (user override)
No Inversion Height Relative Humidity: 54%

SOURCE STRENGTH:

Leak from hole in vertical cylindrical tank
Flammable chemical escaping from tank (not burning)
Tank Diameter: 5.8 meters Tank Length: 12.9 meters
Tank Volume: 341 cubic meters
Tank contains liquid Internal Temperature: 18.54° C
Chemical Mass in Tank: 177,475 kilograms
Tank is 85% full
Circular Opening Diameter: 5 centimeters
Opening is 11.0 meters from tank bottom
Release Duration: ALOHA limited the duration to 1 hour
Max Average Sustained Release Rate: 1,970 kilograms/min
(averaged over a minute or more)
Total Amount Released: 10,222 kilograms
Note: The chemical escaped as a mixture of gas and aerosol (two phase flow).

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